

1 Archetype 01_CH (0_1918)

Swiss CH building assumptions; reference weather ZUESTA, TIME_HORIZON=present, SCENARIO=none, occupancy SIA2024 (residential).

1.1 Geometry inputs

Component	Area [m²]	U-value [W/(m²·K)]	g-value [-]
Floor	95.9	0.9	—
Facade (opaque)	174	0.81	—
Roof	109	0.46	—
Windows	33.1	2	0.6

1.2 Thermal mass

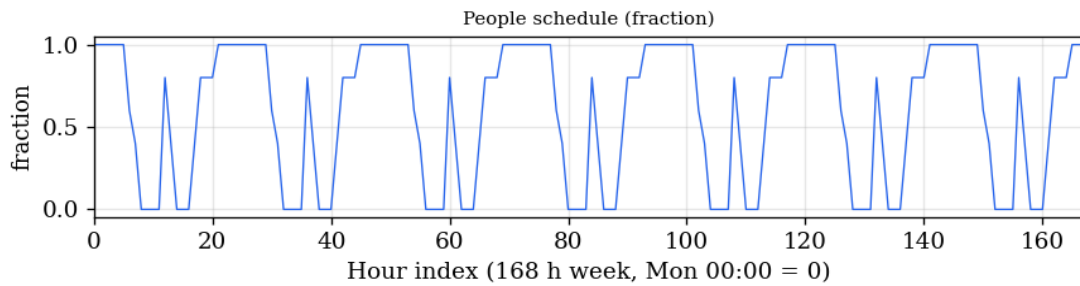
Quantity	Value
Heat capacity class (TABULA / ISO 13790)	heavy
Thermal mass capacitance [kWh/K]	16.11
Thermal mass per floor area [Wh/(m²·K)]	72.22
Conditioned floor area [m²]	223
TABULA building code	AT.N.SFH.01.Gen.ReEx.001.002
TABULA period label	0_1918

1.3 Internal loads

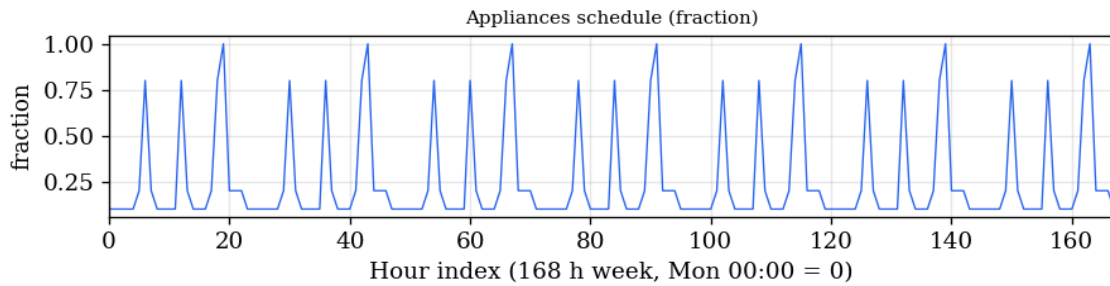
Load	Intensity	Schedule	Seasonal variation
People (sensible internal gains)	1.78 W/m² (0.02 pers/m² × 89.0 W/pers)	Weekly hourly profile (168 h, residential); sensible heat 89.0 W per person	Constant monthly multiplier on weekly profile: 0.8 (SIA presence utilisation)
Appliances (electricity → internal gains)	10 W/m²	Weekly hourly profile (168 h, residential)	Uniform annual multiplier on weekly profile: 0.7994610375028071 (SIA t_A,Ps vs daily equivalent)
Lighting (electricity → internal gains)	3 W/m²	SIA 2024 residential Nutzungsstunden (168 h weekly shape)	Annual full-load target t_L = 846.0 h; Monthly scale 1 + 0.35·cos(2π·m/12) (winter higher)

1.4 Weekly schedule profiles

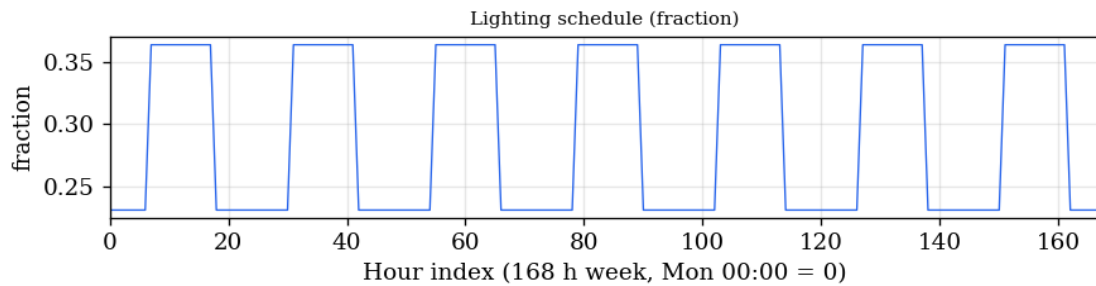
People schedule (weekly)



Appliances schedule (weekly)



Lighting schedule (weekly)



1.5 Ventilation

Mechanism	Parameter	Value / schedule
Infiltration (Swiss, replaces TABULA n _{air} infiltration)	Air change rate	0.3 1/h
Hygienic ventilation (SIA floor area)	Flow per conditioned area	0.6 m ³ /(h·m ²); enabled: yes
Occupancy-driven ventilation	Flow per person (when present)	29 m ³ /(h·person); enabled: yes; schedule follows SIA 2024 people profile
Summer window ventilation	Target ACH when active	2 1/h; enabled: yes
Summer window ventilation	Running-mean outdoor enable threshold [°C]	-10
Summer window ventilation	Open when outdoor cooler than operative	yes

2 Archetype 02_CH (1919_1945)

Swiss CH building assumptions; reference weather ZUESTA, TIME_HORIZON=present, SCENARIO=none, occupancy SIA2024 (residential).

2.1 Geometry inputs

Component	Area [m²]	U-value [W/(m²·K)]	g-value [-]
Floor	81.5	0.9	—
Facade (opaque)	184	0.9	—
Roof	96	0.44	—
Windows	31.2	1.8	0.6

2.2 Thermal mass

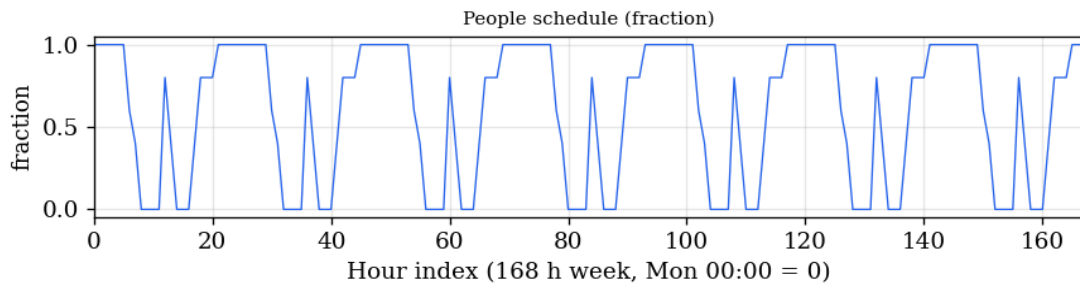
Quantity	Value
Heat capacity class (TABULA / ISO 13790)	medium
Thermal mass capacitance [kWh/K]	8.938
Thermal mass per floor area [Wh/(m²·K)]	45.83
Conditioned floor area [m²]	195
TABULA building code	AT.N.SFH.02.Gen.ReEx.001.002
TABULA period label	1919_1945

2.3 Internal loads

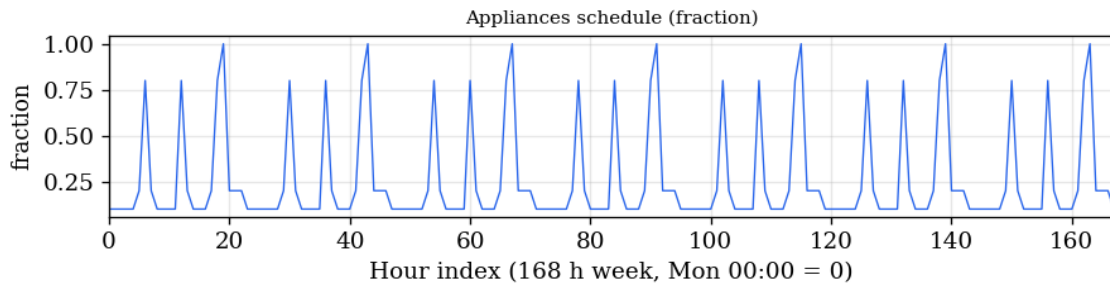
Load	Intensity	Schedule	Seasonal variation
People (sensible internal gains)	1.78 W/m² (0.02 pers/m² × 89.0 W/pers)	Weekly hourly profile (168 h, residential); sensible heat 89.0 W per person	Constant monthly multiplier on weekly profile: 0.8 (SIA presence utilisation)
Appliances (electricity → internal gains)	10 W/m²	Weekly hourly profile (168 h, residential)	Uniform annual multiplier on weekly profile: 0.7994610375028071 (SIA t_A,Ps vs daily equivalent)
Lighting (electricity → internal gains)	3 W/m²	SIA 2024 residential Nutzungsstunden (168 h weekly shape)	Annual full-load target t_L = 846.0 h; Monthly scale 1 + 0.35·cos(2π·m/12) (winter higher)

2.4 Weekly schedule profiles

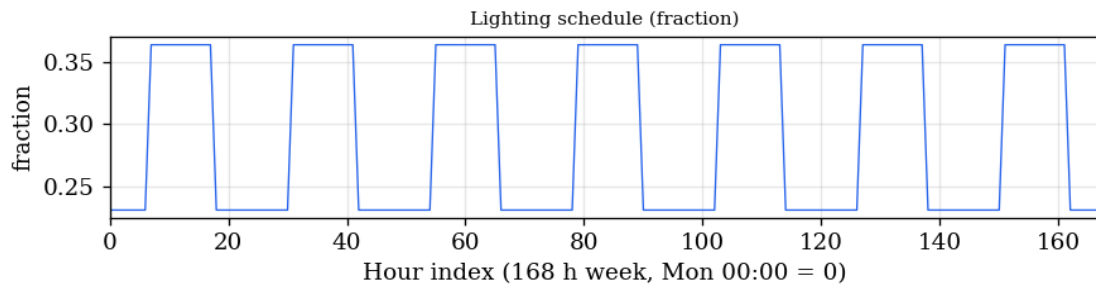
People schedule (weekly)



Appliances schedule (weekly)



Lighting schedule (weekly)



2.5 Ventilation

Mechanism	Parameter	Value / schedule
Infiltration (Swiss, replaces TABULA $n_{\text{air_infiltration}}$)	Air change rate	0.3 1/h
Hygienic ventilation (SIA floor area)	Flow per conditioned area	0.6 m ³ /(h·m ²); enabled: yes
Occupancy-driven ventilation	Flow per person (when present)	29 m ³ /(h·person); enabled: yes; schedule follows SIA 2024 people profile
Summer window ventilation	Target ACH when active	2 1/h; enabled: yes
Summer window ventilation	Running-mean outdoor enable threshold [°C]	-10
Summer window ventilation	Open when outdoor cooler than operative	yes

3 Archetype 03_CH (1946_1960)

Swiss CH building assumptions; reference weather ZUESTA, TIME_HORIZON=present, SCENARIO=none, occupancy SIA2024 (residential).

3.1 Geometry inputs

Component	Area [m²]	U-value [W/(m²·K)]	g-value [-]
Floor	86.1	0.98	—
Facade (opaque)	168	0.81	—
Roof	95.4	0.44	—
Windows	28.8	1.6	0.6

3.2 Thermal mass

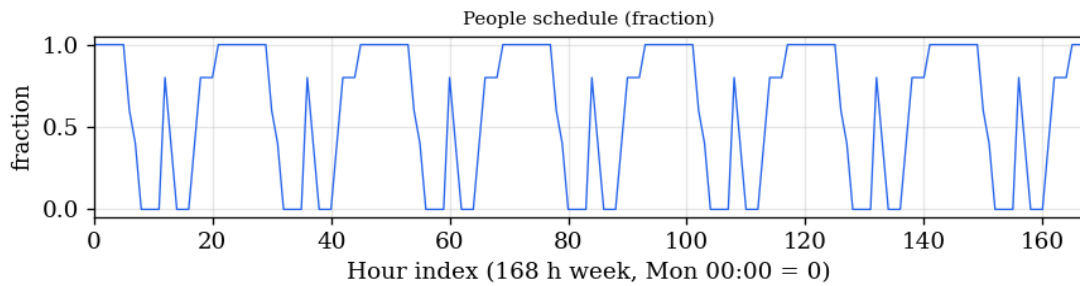
Quantity	Value
Heat capacity class (TABULA / ISO 13790)	medium
Thermal mass capacitance [kWh/K]	8.25
Thermal mass per floor area [Wh/(m²·K)]	45.83
Conditioned floor area [m²]	180
TABULA building code	AT.N.SFH.03.Gen.ReEx.001.002
TABULA period label	1946_1960

3.3 Internal loads

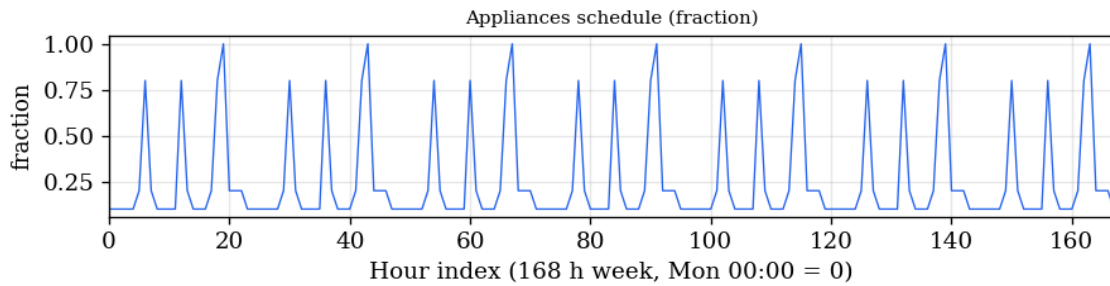
Load	Intensity	Schedule	Seasonal variation
People (sensible internal gains)	1.78 W/m² (0.02 pers/m² × 89.0 W/pers)	Weekly hourly profile (168 h, residential); sensible heat 89.0 W per person	Constant monthly multiplier on weekly profile: 0.8 (SIA presence utilisation)
Appliances (electricity → internal gains)	10 W/m²	Weekly hourly profile (168 h, residential)	Uniform annual multiplier on weekly profile: 0.7994610375028071 (SIA t_A,Ps vs daily equivalent)
Lighting (electricity → internal gains)	3 W/m²	SIA 2024 residential Nutzungsstunden (168 h weekly shape)	Annual full-load target t_L = 846.0 h; Monthly scale 1 + 0.35·cos(2π·m/12) (winter higher)

3.4 Weekly schedule profiles

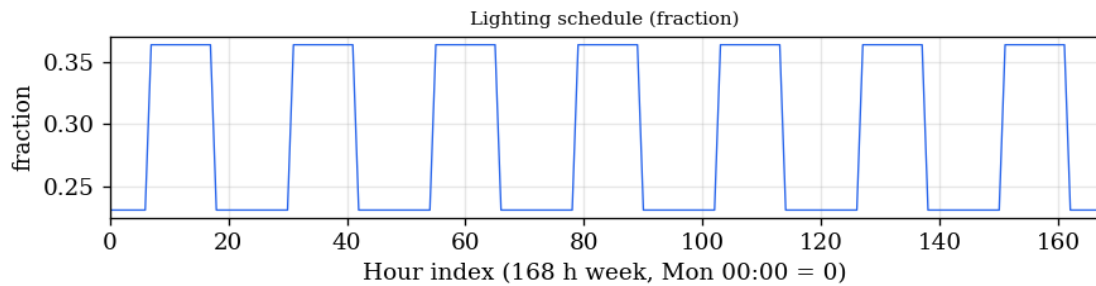
People schedule (weekly)



Appliances schedule (weekly)



Lighting schedule (weekly)



3.5 Ventilation

Mechanism	Parameter	Value / schedule
Infiltration (Swiss, replaces TABULA n _{air} infiltration)	Air change rate	0.3 1/h
Hygienic ventilation (SIA floor area)	Flow per conditioned area	0.6 m ³ /(h·m ²); enabled: yes
Occupancy-driven ventilation	Flow per person (when present)	29 m ³ /(h·person); enabled: yes; schedule follows SIA 2024 people profile
Summer window ventilation	Target ACH when active	2 1/h; enabled: yes
Summer window ventilation	Running-mean outdoor enable threshold [°C]	-10
Summer window ventilation	Open when outdoor cooler than operative	yes

4 Archetype 04_CH (1961_1970)

Swiss CH building assumptions; reference weather ZUESTA, TIME_HORIZON=present, SCENARIO=none, occupancy SIA2024 (residential).

4.1 Geometry inputs

Component	Area [m²]	U-value [W/(m²·K)]	g-value [-]
Floor	99.2	1	—
Facade (opaque)	163	0.7	—
Roof	109	0.47	—
Windows	30.4	1.7	0.6

4.2 Thermal mass

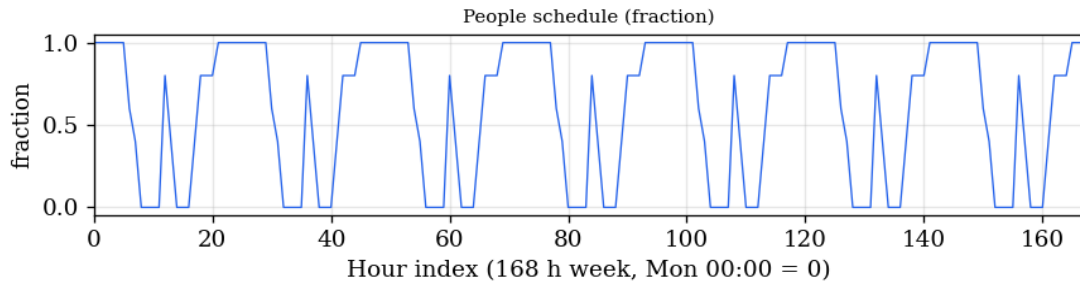
Quantity	Value
Heat capacity class (TABULA / ISO 13790)	heavy
Thermal mass capacitance [kWh/K]	13.72
Thermal mass per floor area [Wh/(m²·K)]	72.22
Conditioned floor area [m²]	190
TABULA building code	AT.N.SFH.04.Gen.ReEx.001.002
TABULA period label	1961_1970

4.3 Internal loads

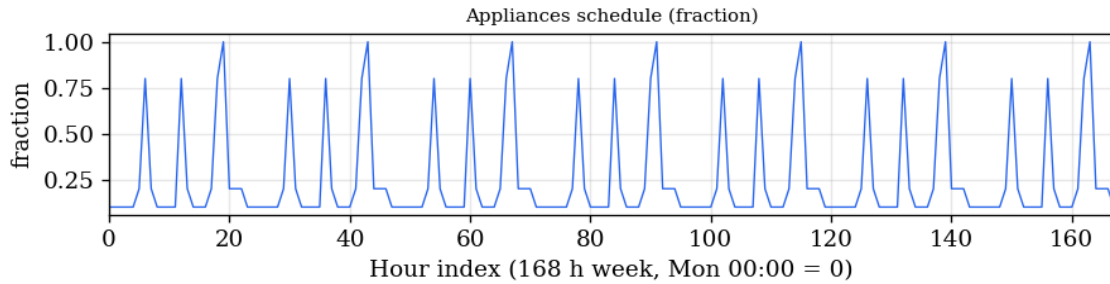
Load	Intensity	Schedule	Seasonal variation
People (sensible internal gains)	1.78 W/m² (0.02 pers/m² × 89.0 W/pers)	Weekly hourly profile (168 h, residential); sensible heat 89.0 W per person	Constant monthly multiplier on weekly profile: 0.8 (SIA presence utilisation)
Appliances (electricity → internal gains)	10 W/m²	Weekly hourly profile (168 h, residential)	Uniform annual multiplier on weekly profile: 0.7994610375028071 (SIA t_A,Ps vs daily equivalent)
Lighting (electricity → internal gains)	3 W/m²	SIA 2024 residential Nutzungsstunden (168 h weekly shape)	Annual full-load target t_L = 846.0 h; Monthly scale 1 + 0.35·cos(2π·m/12) (winter higher)

4.4 Weekly schedule profiles

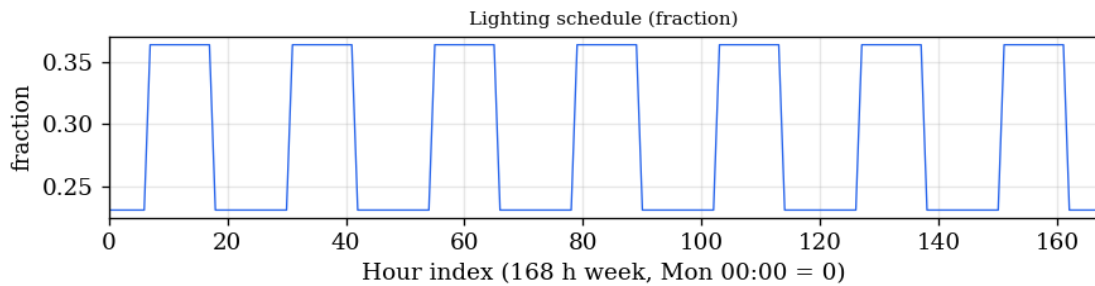
People schedule (weekly)



Appliances schedule (weekly)



Lighting schedule (weekly)



4.5 Ventilation

Mechanism	Parameter	Value / schedule
Infiltration (Swiss, replaces TABULA n _{air} infiltration)	Air change rate	0.3 1/h
Hygienic ventilation (SIA floor area)	Flow per conditioned area	0.6 m ³ /(h·m ²); enabled: yes
Occupancy-driven ventilation	Flow per person (when present)	29 m ³ /(h·person); enabled: yes; schedule follows SIA 2024 people profile
Summer window ventilation	Target ACH when active	2 1/h; enabled: yes
Summer window ventilation	Running-mean outdoor enable threshold [°C]	-10
Summer window ventilation	Open when outdoor cooler than operative	yes

5 Archetype 05_CH (1971_1980)

Swiss CH building assumptions; reference weather ZUESTA, TIME_HORIZON=present, SCENARIO=none, occupancy SIA2024 (residential).

5.1 Geometry inputs

Component	Area [m²]	U-value [W/(m²·K)]	g-value [-]
Floor	103	0.88	—
Facade (opaque)	163	0.56	—
Roof	115	0.41	—
Windows	32	1.7	0.6

5.2 Thermal mass

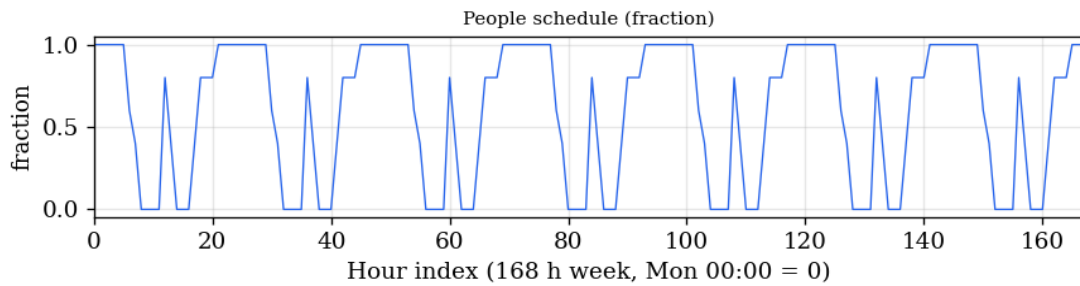
Quantity	Value
Heat capacity class (TABULA / ISO 13790)	medium
Thermal mass capacitance [kWh/K]	9.167
Thermal mass per floor area [Wh/(m²·K)]	45.83
Conditioned floor area [m²]	200
TABULA building code	AT.N.SFH.04.Gen.ReEx.001.002
TABULA period label	1971_1980

5.3 Internal loads

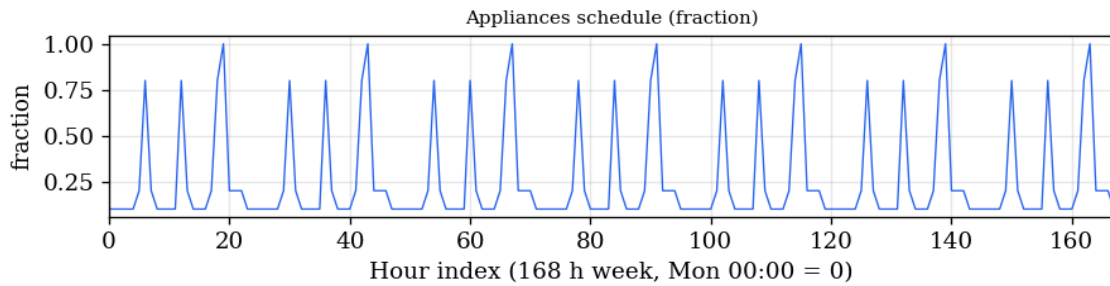
Load	Intensity	Schedule	Seasonal variation
People (sensible internal gains)	1.78 W/m² (0.02 pers/m² × 89.0 W/pers)	Weekly hourly profile (168 h, residential); sensible heat 89.0 W per person	Constant monthly multiplier on weekly profile: 0.8 (SIA presence utilisation)
Appliances (electricity → internal gains)	10 W/m²	Weekly hourly profile (168 h, residential)	Uniform annual multiplier on weekly profile: 0.7994610375028071 (SIA t_A,Ps vs daily equivalent)
Lighting (electricity → internal gains)	3 W/m²	SIA 2024 residential Nutzungsstunden (168 h weekly shape)	Annual full-load target t_L = 846.0 h; Monthly scale 1 + 0.35·cos(2π·m/12) (winter higher)

5.4 Weekly schedule profiles

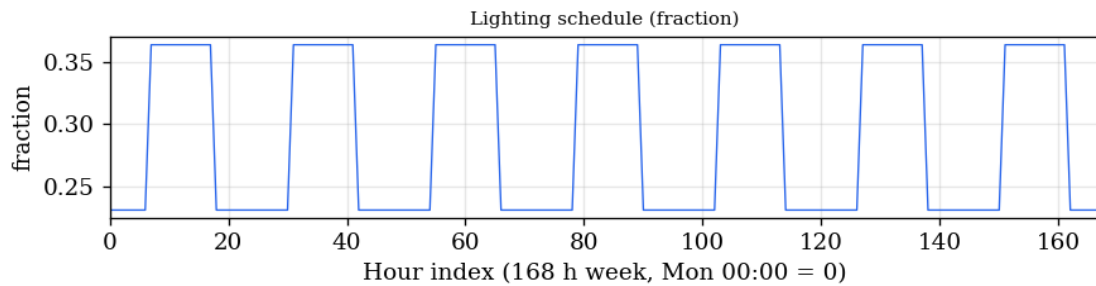
People schedule (weekly)



Appliances schedule (weekly)



Lighting schedule (weekly)



5.5 Ventilation

Mechanism	Parameter	Value / schedule
Infiltration (Swiss, replaces TABULA n _{air} infiltration)	Air change rate	0.3 1/h
Hygienic ventilation (SIA floor area)	Flow per conditioned area	0.6 m ³ /(h·m ²); enabled: yes
Occupancy-driven ventilation	Flow per person (when present)	29 m ³ /(h·person); enabled: yes; schedule follows SIA 2024 people profile
Summer window ventilation	Target ACH when active	2 1/h; enabled: yes
Summer window ventilation	Running-mean outdoor enable threshold [°C]	-10
Summer window ventilation	Open when outdoor cooler than operative	yes

6 Archetype 06_CH (1981_1990)

Swiss CH building assumptions; reference weather ZUESTA, TIME_HORIZON=present, SCENARIO=none, occupancy SIA2024 (residential).

6.1 Geometry inputs

Component	Area [m²]	U-value [W/(m²·K)]	g-value [-]
Floor	88.2	0.61	—
Facade (opaque)	160	0.4	—
Roof	103	0.36	—
Windows	31.7	1.9	0.6

6.2 Thermal mass

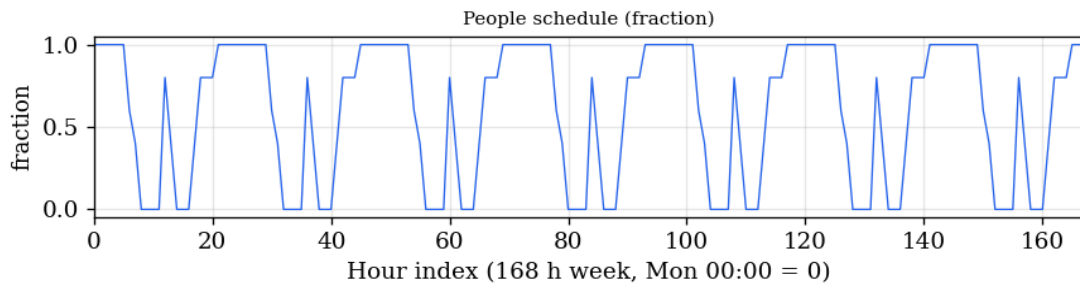
Quantity	Value
Heat capacity class (TABULA / ISO 13790)	medium
Thermal mass capacitance [kWh/K]	9.075
Thermal mass per floor area [Wh/(m²·K)]	45.83
Conditioned floor area [m²]	198
TABULA building code	AT.N.SFH.05.Gen.ReEx.001.002
TABULA period label	1981_1990

6.3 Internal loads

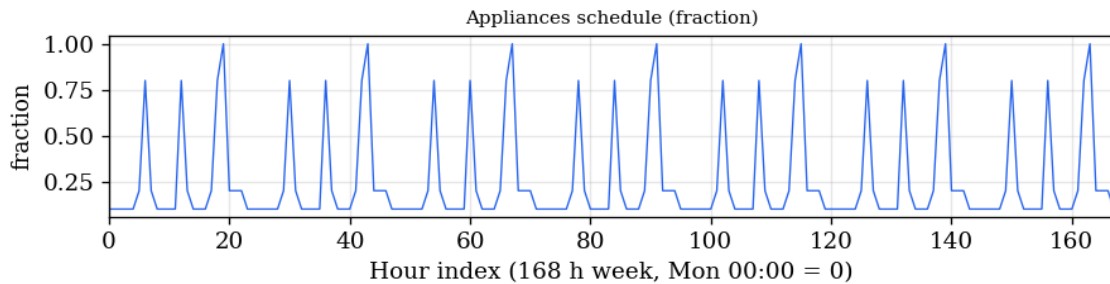
Load	Intensity	Schedule	Seasonal variation
People (sensible internal gains)	1.78 W/m² (0.02 pers/m² × 89.0 W/pers)	Weekly hourly profile (168 h, residential); sensible heat 89.0 W per person	Constant monthly multiplier on weekly profile: 0.8 (SIA presence utilisation)
Appliances (electricity → internal gains)	10 W/m²	Weekly hourly profile (168 h, residential)	Uniform annual multiplier on weekly profile: 0.7994610375028071 (SIA t_A,Ps vs daily equivalent)
Lighting (electricity → internal gains)	3 W/m²	SIA 2024 residential Nutzungsstunden (168 h weekly shape)	Annual full-load target t_L = 846.0 h; Monthly scale 1 + 0.35·cos(2π·m/12) (winter higher)

6.4 Weekly schedule profiles

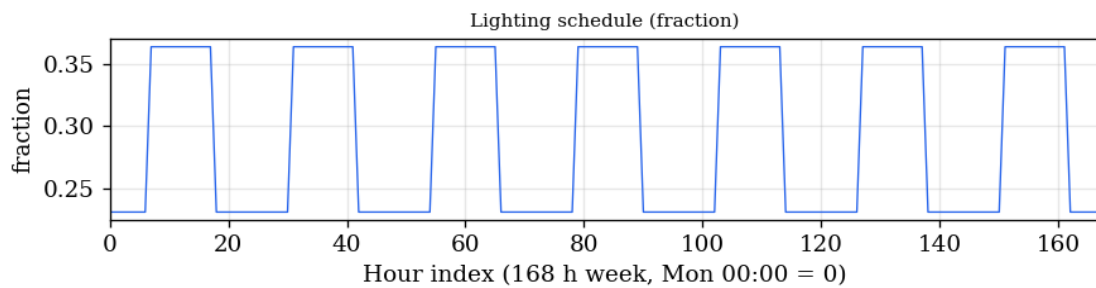
People schedule (weekly)



Appliances schedule (weekly)



Lighting schedule (weekly)



6.5 Ventilation

Mechanism	Parameter	Value / schedule
Infiltration (Swiss, replaces TABULA $n_{\text{air_infiltration}}$)	Air change rate	0.15 1/h
Hygienic ventilation (SIA floor area)	Flow per conditioned area	0.6 m ³ /(h·m ²); enabled: yes
Occupancy-driven ventilation	Flow per person (when present)	29 m ³ /(h·person); enabled: yes; schedule follows SIA 2024 people profile
Summer window ventilation	Target ACH when active	2 1/h; enabled: yes
Summer window ventilation	Running-mean outdoor enable threshold [°C]	-10
Summer window ventilation	Open when outdoor cooler than operative	yes

7 Archetype 07_CH (1991_2000)

Swiss CH building assumptions; reference weather ZUESTA, TIME_HORIZON=present, SCENARIO=none, occupancy SIA2024 (residential).

7.1 Geometry inputs

Component	Area [m²]	U-value [W/(m²·K)]	g-value [-]
Floor	81	0.55	—
Facade (opaque)	164	0.35	—
Roof	96	0.34	—
Windows	31.8	1.7	0.6

7.2 Thermal mass

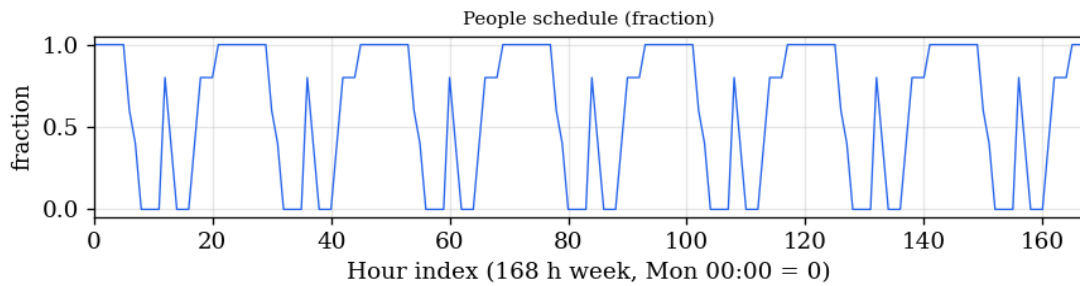
Quantity	Value
Heat capacity class (TABULA / ISO 13790)	heavy
Thermal mass capacitance [kWh/K]	14.37
Thermal mass per floor area [Wh/(m²·K)]	72.22
Conditioned floor area [m²]	199
TABULA building code	AT.N.SFH.06.Gen.ReEx.001.002
TABULA period label	1991_2000

7.3 Internal loads

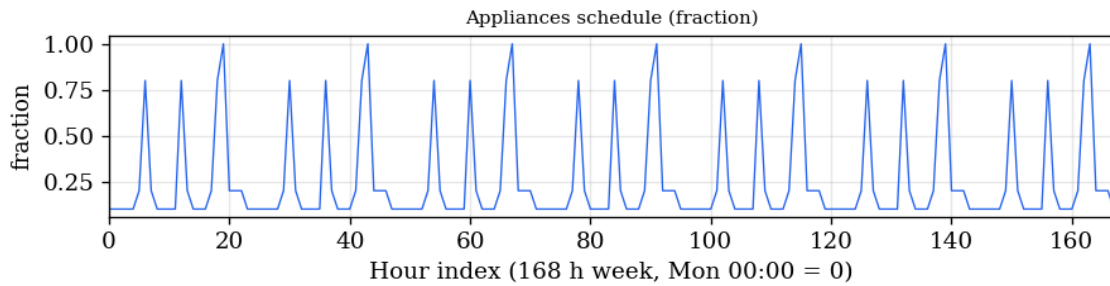
Load	Intensity	Schedule	Seasonal variation
People (sensible internal gains)	1.78 W/m² (0.02 pers/m² × 89.0 W/pers)	Weekly hourly profile (168 h, residential); sensible heat 89.0 W per person	Constant monthly multiplier on weekly profile: 0.8 (SIA presence utilisation)
Appliances (electricity → internal gains)	10 W/m²	Weekly hourly profile (168 h, residential)	Uniform annual multiplier on weekly profile: 0.7994610375028071 (SIA t_A,Ps vs daily equivalent)
Lighting (electricity → internal gains)	3 W/m²	SIA 2024 residential Nutzungsstunden (168 h weekly shape)	Annual full-load target t_L = 846.0 h; Monthly scale 1 + 0.35·cos(2π·m/12) (winter higher)

7.4 Weekly schedule profiles

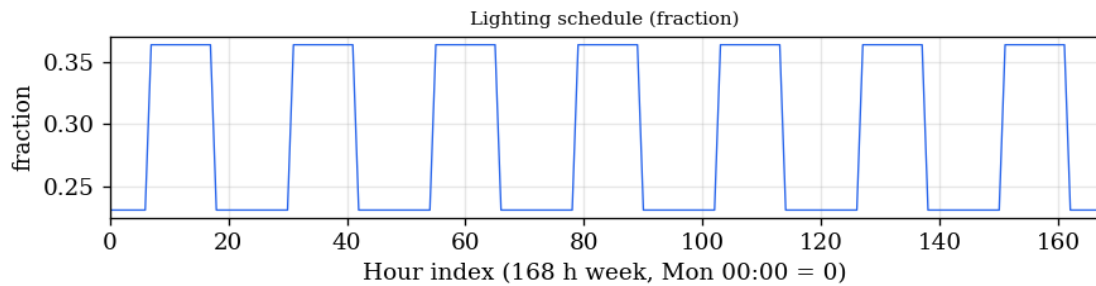
People schedule (weekly)



Appliances schedule (weekly)



Lighting schedule (weekly)



7.5 Ventilation

Mechanism	Parameter	Value / schedule
Infiltration (Swiss, replaces TABULA n _{air} infiltration)	Air change rate	0.15 1/h
Hygienic ventilation (SIA floor area)	Flow per conditioned area	0.6 m ³ /(h·m ²); enabled: yes
Occupancy-driven ventilation	Flow per person (when present)	29 m ³ /(h·person); enabled: yes; schedule follows SIA 2024 people profile
Summer window ventilation	Target ACH when active	2 1/h; enabled: yes
Summer window ventilation	Running-mean outdoor enable threshold [°C]	-10
Summer window ventilation	Open when outdoor cooler than operative	yes

8 Archetype 08_CH (2001_2010)

Swiss CH building assumptions; reference weather ZUESTA, TIME_HORIZON=present, SCENARIO=none, occupancy SIA2024 (residential).

8.1 Geometry inputs

Component	Area [m²]	U-value [W/(m²·K)]	g-value [-]
Floor	84.2	0.38	—
Facade (opaque)	178	0.3	—
Roof	96.8	0.26	—
Windows	32.2	1.5	0.6

8.2 Thermal mass

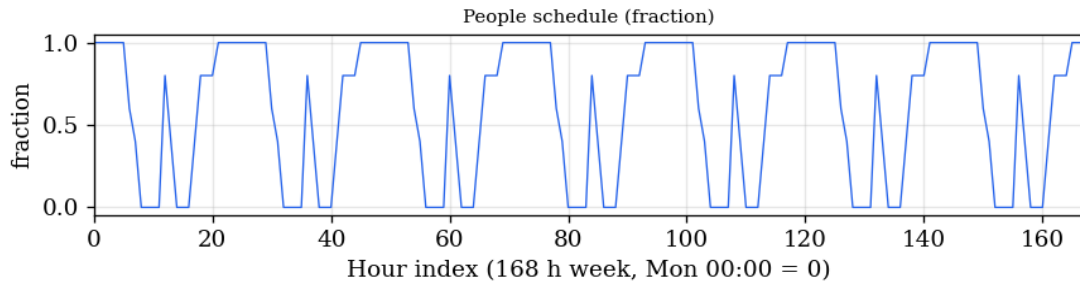
Quantity	Value
Heat capacity class (TABULA / ISO 13790)	heavy
Thermal mass capacitance [kWh/K]	14.52
Thermal mass per floor area [Wh/(m²·K)]	72.22
Conditioned floor area [m²]	201
TABULA building code	AT.N.SFH.07.Gen.ReEx.001.002
TABULA period label	2001_2010

8.3 Internal loads

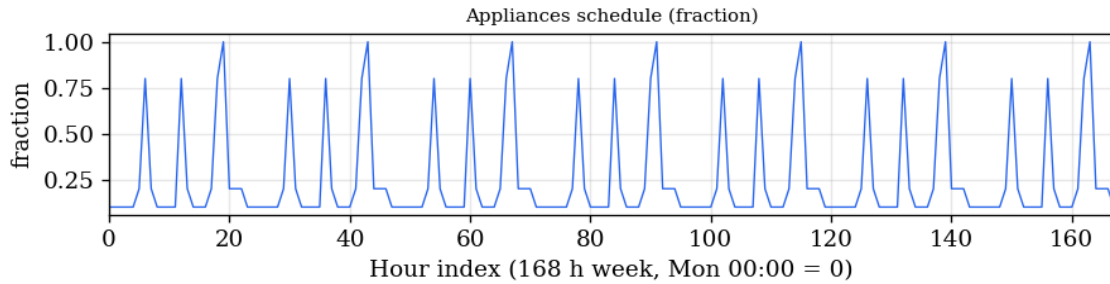
Load	Intensity	Schedule	Seasonal variation
People (sensible internal gains)	1.78 W/m² (0.02 pers/m² × 89.0 W/pers)	Weekly hourly profile (168 h, residential); sensible heat 89.0 W per person	Constant monthly multiplier on weekly profile: 0.8 (SIA presence utilisation)
Appliances (electricity → internal gains)	10 W/m²	Weekly hourly profile (168 h, residential)	Uniform annual multiplier on weekly profile: 0.7994610375028071 (SIA t_A,Ps vs daily equivalent)
Lighting (electricity → internal gains)	3 W/m²	SIA 2024 residential Nutzungsstunden (168 h weekly shape)	Annual full-load target t_L = 846.0 h; Monthly scale 1 + 0.35·cos(2π·m/12) (winter higher)

8.4 Weekly schedule profiles

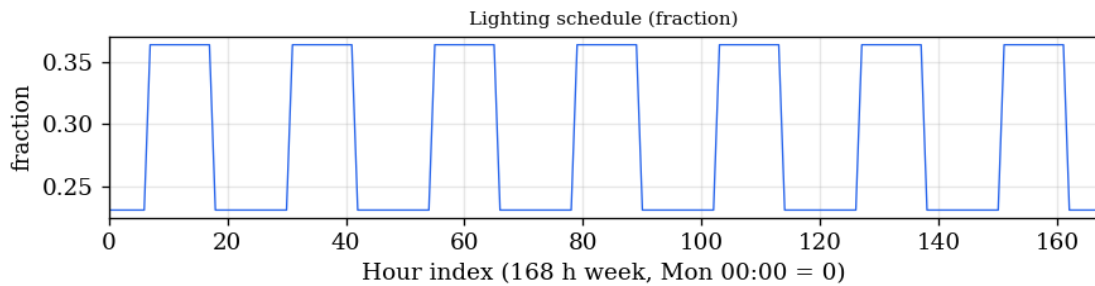
People schedule (weekly)



Appliances schedule (weekly)



Lighting schedule (weekly)



8.5 Ventilation

Mechanism	Parameter	Value / schedule
Infiltration (Swiss, replaces TABULA n _{air} infiltration)	Air change rate	0.15 1/h
Hygienic ventilation (SIA floor area)	Flow per conditioned area	0.6 m ³ /(h·m ²); enabled: yes
Occupancy-driven ventilation	Flow per person (when present)	29 m ³ /(h·person); enabled: yes; schedule follows SIA 2024 people profile
Summer window ventilation	Target ACH when active	2 1/h; enabled: yes
Summer window ventilation	Running-mean outdoor enable threshold [°C]	-10
Summer window ventilation	Open when outdoor cooler than operative	yes

9 Archetype 09_CH (2011_2019)

Swiss CH building assumptions; reference weather ZUESTA, TIME_HORIZON=present, SCENARIO=none, occupancy SIA2024 (residential).

9.1 Geometry inputs

Component	Area [m²]	U-value [W/(m²·K)]	g-value [-]
Floor	93.9	0.21	—
Facade (opaque)	181	0.17	—
Roof	100	0.17	—
Windows	37.3	1	0.65

9.2 Thermal mass

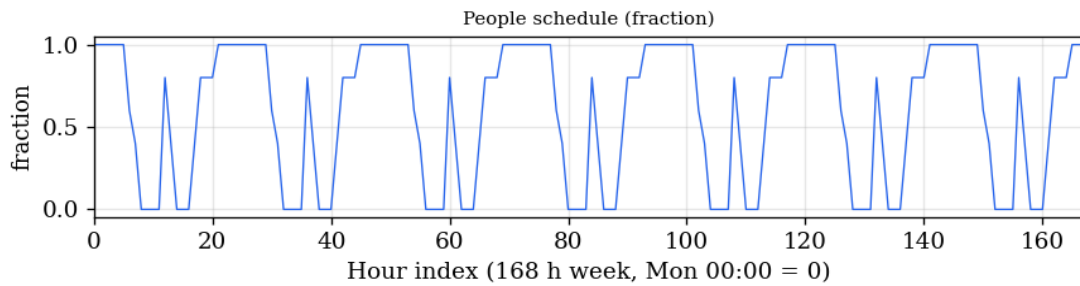
Quantity	Value
Heat capacity class (TABULA / ISO 13790)	heavy
Thermal mass capacitance [kWh/K]	14.88
Thermal mass per floor area [Wh/(m²·K)]	72.22
Conditioned floor area [m²]	206
TABULA building code	AT.N.SFH.08.Gen.ReEx.001.002
TABULA period label	2011_2019

9.3 Internal loads

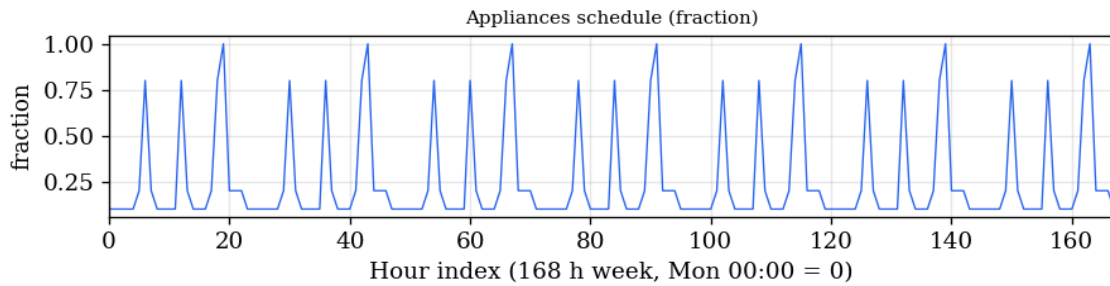
Load	Intensity	Schedule	Seasonal variation
People (sensible internal gains)	1.78 W/m² (0.02 pers/m² × 89.0 W/pers)	Weekly hourly profile (168 h, residential); sensible heat 89.0 W per person	Constant monthly multiplier on weekly profile: 0.8 (SIA presence utilisation)
Appliances (electricity → internal gains)	10 W/m²	Weekly hourly profile (168 h, residential)	Uniform annual multiplier on weekly profile: 0.7994610375028071 (SIA t_A,Ps vs daily equivalent)
Lighting (electricity → internal gains)	3 W/m²	SIA 2024 residential Nutzungsstunden (168 h weekly shape)	Annual full-load target t_L = 846.0 h; Monthly scale 1 + 0.35·cos(2π·m/12) (winter higher)

9.4 Weekly schedule profiles

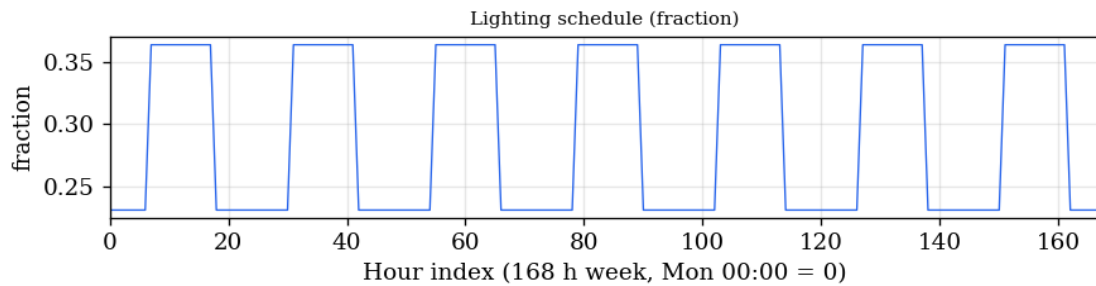
People schedule (weekly)



Appliances schedule (weekly)



Lighting schedule (weekly)



9.5 Ventilation

Mechanism	Parameter	Value / schedule
Infiltration (Swiss, replaces TABULA n _{air} infiltration)	Air change rate	0.15 1/h
Hygienic ventilation (SIA floor area)	Flow per conditioned area	0.6 m ³ /(h·m ²); enabled: yes
Occupancy-driven ventilation	Flow per person (when present)	29 m ³ /(h·person); enabled: yes; schedule follows SIA 2024 people profile
Summer window ventilation	Target ACH when active	2 1/h; enabled: yes
Summer window ventilation	Running-mean outdoor enable threshold [°C]	-10
Summer window ventilation	Open when outdoor cooler than operative	yes